

Agenda

1.) Announcements

A.) HW 7.3, 7.4, 7.5 due today

B.) Discussion Quiz 3 tomorrow over

2.) Finish Section 7.8

3.) Start Section 9.1

Improper Integrals on Discontinuous Integrals

Definition: Let $f(x)$ be a function.

1.) If $f(x)$ is continuous on $[a, b)$ and discontinuous at $x=b$, then

$$\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$$

2.) If $f(x)$ is continuous on $(a, b]$ and discontinuous at $x=a$, then

$$\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx$$

We say the improper integral converges if it exists.
Diverges if it DNE

improper @ 3
 $3-3=0$

Example:

$$\begin{aligned} \int_3^7 \frac{dx}{\sqrt{x-3}} &= \lim_{t \rightarrow 3^+} \int_t^7 (x-3)^{-1/2} dx \\ &= \lim_{t \rightarrow 3^+} \left[2(x-3)^{1/2} \right]_t^7 \\ &= \lim_{t \rightarrow 3^+} \left[2\sqrt{7-3} - 2\sqrt{t-3} \right] \\ &= 4 - 0 \Rightarrow \textcircled{1} \text{ converges} \end{aligned}$$

Activity: $\int_1^2 \frac{4x}{\sqrt[3]{x^2-4}} dx$ improper
② $u = x^2 - 4$
 $du = 2x$

$$= \lim_{t \rightarrow 2^-} \int_1^t 4x (x^2-4)^{-1/3} dx$$

$$= \lim_{t \rightarrow 2^-} \int_{-3}^{t^2-4} 2u^{-1/3} du$$

$$= \lim_{t \rightarrow 2^-} \left[3u^{2/3} \right]_{-3}^{t^2-4}$$

$$= \lim_{t \rightarrow 2^-} \left[3(t^2-4)^{2/3} - 3(-3)^{2/3} \right] = \underbrace{-3(-3)^{2/3}}_{\text{converges}}$$

as $t \rightarrow 2$

If f is ctns. on $[a, b]$ except some c in (a, b)
 then $\int_a^b f(x) dx \rightarrow \int_a^c f(x) dx + \int_c^b f(x) dx$

discont. @ $x=1$

Example: $\int_0^2 \frac{3x^2}{x^3-1} dx$

$u = x^3 - 1$
 $du = 3x^2 dx$

$$\int_0^1 \frac{3x^2}{x^3-1} dx = \lim_{t \rightarrow 1^-} \int_0^t \frac{3x^2}{x^3-1} dx$$

$$= \lim_{t \rightarrow 1^-} \int_{-1}^{t^3-1} \frac{du}{u}$$

$$= \lim_{t \rightarrow 1^-} \left[\ln|u| \right]_{-1}^{t^3-1}$$

$$= \lim_{t \rightarrow 1^-} \left[\ln|t^3-1| - \ln|-1| \right] = \underbrace{-\infty}_{\text{diverges}}$$

no first one

because the mistakes
charges, you do not
have to check the work

Section 9.1 - Models for Differential Equations

Differential Equations

Suppose that we want to find functions y where

① $\frac{dy}{dx} = 3x^2 - 16x + 2$ OR ② $\frac{dy}{dx} = y$

For diff eq like #1, we can use the FTC.

① $y = x^3 - 8x^2 + 2x + C$
(read: eqn -) just x's)

② $\frac{dy}{dx} = y$???

don't have tools but
can make educated
guesses about
specific solutions

$y = e^x$
 $y = -5e^x$
 $y = 0$

can infer the
general

solution of

$\frac{dy}{dx} = y$ is

$y = Ce^x$ (C is a constant)

Definition: A differential equation is an equation which contains an unknown function ^{typically y} and one or more of its derivatives. The order of a differential equation is its highest-order derivative.

Example: Identify the order of the differential equation

① $\frac{dy}{dx} = 3x^2 + 2xy - y$

first order

② $3y^2 - 2y \frac{dy}{dx} + \frac{d^2y}{dx^2} = e^x$

second order

Modeling Populations

Example: Suppose P is the population of an organism as a function of time. Let its birth rate be 5% and its death rate be 2%.

$$\frac{dP}{dt} = .05P - .02P$$

$$\frac{dP}{dt} = .03P$$

$$P = Ce^{.03t}$$

$$P(0) = C$$

Solutions to Differential Equations

We call a function a solution if it satisfies the given differential equation.

Example: Show that $y = 2e^{-x} + x - 1$ is a solution to $y' = x - y$

Activity: Show that $y = 4(x^2 - 1)^{-1/2}$ is a solution to $(x^2 - 1)y' + xy = 0$.

Example: Find all values of k such that $y = \sin(kt)$ is a solution to $y'' + 16y = 0$

Activity: Find the values of n so that $y = e^{nt}$ is a solution to $y' = 6y - y''$